

Modulation of inflammatory genes by natural dietary bioactive compounds.

Several epidemiologic studies have shown that chronic inflammation predisposes individuals to various types of cancer. Many cancers arise from sites of infection, chronic irritation, and inflammation. Conversely, an oncogenic change induces an inflammatory microenvironment that promotes the development of tumors. Natural bioactive compounds in dietary plant products including fruits, vegetables, grains, legumes, tea, and wine are claimed to help prevent cancer, degenerative diseases, and chronic and acute inflammation. Modern methods in cell and molecular biology allow us to understand the interactions of different natural bioactive compounds with basic mechanisms of inflammatory response. The molecular pathways of this cancer-related inflammation are now unraveled. Natural bioactive compounds exert anti-inflammatory activity by modulating pro-inflammatory gene expressions have shown promising chemopreventive activity. This review summarizes current knowledge on natural bioactive compounds that act through the signaling pathways and modulate inflammatory gene expressions, thus providing evidence for these substances in cancer chemopreventive action.